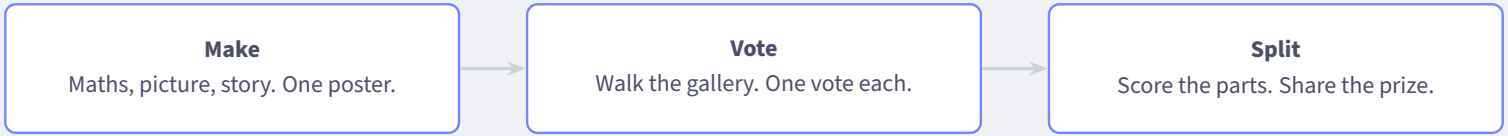
Cooperative games

Three people made one thing. Now split the prize.



Winning team:

Votes:

I will cover the poster and reveal the parts in combination. Score what is on show out of ten.

Maths	Picture	Story	The whole aid **10**
Maths + picture	Maths + story	Picture + story	

Those seven numbers are the characteristic function v. Anchor the whole aid at ten and judge the rest against it.

While you play

In threes, for twelve minutes: the Theorist writes the maths, the Artist draws the picture, the Storyteller tells the story. One poster, three clearly separated parts. Then everybody votes once, for any aid but their own.

Six ways the aid could have come together

Order	Maths adds	Picture adds	Story adds
Maths, picture, story			
Maths, story, picture			
Picture, maths, story			
Picture, story, maths			
Story, maths, picture			
Story, picture, maths			
Average			

Maths ____ Picture ____ Story ____

They should add to ____, which is the whole value of the aid. That property is called _____.

The three parts on their own are worth ____ + ____ + ____ = ____, so ____ of the aid's value came from the parts _____.

For each person, how much more did the Shapley value give them than their part was worth alone?

Maths Picture Story

The part that gains most is the one that _____, not the one that is best on its own.

What we just did in the room is **Question 1** on the Cooperative Games page, with the class's scores replaced by fixed ones.

(a)	Define a characteristic function game and the Shapley value.	[4]
(b)	Marginal contributions in each of the six orderings.	[5]
(c)	Hence the Shapley value.	[2]
(d)	Verify efficiency, and explain how the synergy is shared.	[4]

vknight.org/gt/topics/cooperative-games.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Scoring every combination of parts	**18.2.1**	Cooperative Games: Definition: characteristic function game
The six orders of arrival	**18.2.9, 18.2.10**	Cooperative Games: Predecessors, Marginal Contribution
Averaging the contributions	**18.2.11**	Cooperative Games: Definition: Shapley Value
Why that split is the fair one	**18.2.5 to 18.2.8**	Cooperative Games: Efficiency, Null Player, Symmetry, Additivity

vknight.org/gtb/chapters/cooperative-games

Quick quiz before you go: vknight.org/gt/topics/cooperative-games.html

Before the next class: read the Cooperative Games chapter as far as the Shapley value, and attempt Question 1.